

Data Analytics Capstone

Topic

Determinants of Urban Air Quality in Indian Cities: Evaluating the Effects of Smart City Status, Electric Vehicle Adoption, Green Cover, and Population Density Using Statistical and Machine Learning Approaches

Final Report

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Abstract

Urban air pollution is a major sustainability and public policy concern in India, particularly in rapidly growing cities where transportation, urbanization, land use, and environmental conditions interact. Fine particulate matter (PM_{2.5}) is an important indicator of urban air quality because of its adverse environmental and public health implications. This study examines the statistical relationships between urban PM_{2.5} concentrations and four selected factors: Smart City status, electric vehicle (EV) adoption, green cover, and population density. The objective is to determine whether these factors are significantly associated with PM_{2.5} levels while accounting for relevant meteorological conditions.

The study uses secondary data obtained from publicly available sources, including Indian government and environmental data repositories. Data from multiple sources were standardized by city and year before integration. The final analytical dataset contains 191 city-year observations covering the period 2017–2025. PM_{2.5} is the dependent variable, while Smart City status, EV adoption, green cover, and population density represent the principal explanatory variables. Temperature, humidity, wind speed, and rainfall are incorporated as meteorological control variables where available. Data preprocessing included data-type validation, duplicate checking, missing-value assessment, consistency checks, and alignment of reporting years.

Two complementary analytical approaches are used. Statistical methods—including Welch's independent samples t-test, Pearson correlation, and multiple linear regression—are used for hypothesis testing and interpretation of relationships. Random Forest Regression and XGBoost Regression are used separately as predictive machine learning approaches to evaluate whether nonlinear models can improve PM_{2.5} prediction. The results show no statistically significant difference in PM_{2.5} between Smart City and non-Smart City

observations, while green cover exhibits a stronger negative bivariate association with PM2.5. EV adoption demonstrates a negligible bivariate association, and population density shows a weaker positive relationship that requires multivariate evaluation. Machine learning results indicate substantially better predictive performance than the ordinary least squares baseline; however, final model selection requires robust cross-validation and identical evaluation conditions across models.

The study is intended to provide evidence that can support urban planners, policymakers, municipal authorities, and environmental agencies in understanding the factors associated with urban PM2.5. Because the study is observational and constrained by publicly available data, the findings are interpreted as statistical associations and predictive relationships rather than causal effects. The final analysis will strengthen model validation, regression diagnostics, and machine learning interpretability before drawing policy recommendations.

Introduction

Background and Context

Rapid urbanization has created substantial environmental and sustainability challenges for Indian cities. Increasing population concentration, transportation demand, infrastructure development, and economic activity can contribute to deteriorating urban air quality. Among the different air pollutants, fine particulate matter (PM2.5) is particularly important because of its small particle size and associated health risks. The interim analysis for this study identifies PM2.5 as the primary outcome variable for evaluating urban air quality and notes substantial variation in PM2.5 levels across the selected Indian cities and years.

Air quality is influenced by multiple interacting environmental, demographic, and policy-related factors. Urban population density may be associated with greater transportation

activity and anthropogenic emissions, while green cover may contribute to pollutant removal and environmental regulation. Government interventions have also increasingly emphasized sustainable urban development. India's Smart Cities Mission seeks to promote technology-enabled infrastructure and sustainable urban development, while electric mobility policies aim to encourage a transition toward cleaner transportation. The study therefore examines whether Smart City status and EV adoption are statistically associated with urban PM_{2.5} levels, alongside ecological and demographic factors.

The study focuses on four principal explanatory variables: Smart City status, EV adoption, green cover, and population density. These variables represent four distinct dimensions of urban sustainability: government-led urban development, transportation transition, ecological conditions, and urban density. Meteorological variables—including temperature, humidity, wind speed, and rainfall—are incorporated as control variables where available because weather conditions can influence pollutant concentration and dispersion. Previous work identified in the project literature also supports the importance of meteorological conditions when modelling PM_{2.5}.

The primary stakeholders for this study include policymakers, urban planners, municipal corporations, the Ministry of Housing and Urban Affairs, the Ministry of Heavy Industries, the Central Pollution Control Board (CPCB), state pollution control boards, and researchers working in sustainable urban development. These stakeholders may benefit from empirical evidence concerning the relationships between urban development characteristics and PM_{2.5}. The study is therefore positioned as a policy-oriented data analytics project rather than solely as a prediction exercise.

It is important, however, to recognize that PM_{2.5} concentrations are influenced by many factors beyond those included in this study. Industrial emissions, construction activity,

biomass burning, traffic congestion, and other emission sources may contribute substantially to urban air pollution. These variables are not included because consistent city-level data covering the selected locations and years were not publicly available. Consequently, the study does not attempt to explain all sources of PM_{2.5} and its findings are limited to the variables represented in the integrated dataset.

Problem Statement

The objective of this study is to **examine the relationship between urban PM_{2.5} concentration and Smart City status, EV adoption, green cover, and population density across selected Indian cities during 2017–2025, while accounting for available meteorological conditions**. The study evaluates statistical relationships using hypothesis testing and multiple linear regression and evaluates predictive performance using regression-based machine learning models. Success of the predictive component is evaluated using **R², RMSE, and MAE**, while the inferential component is evaluated at a **5% significance level**.

Despite substantial government attention to Smart City development and electric mobility, empirical evidence regarding their relationship with urban PM_{2.5} remains limited when these policy variables are examined together with ecological and demographic characteristics. Existing studies frequently focus on individual determinants of air pollution, whereas comparable city-level datasets integrating policy, transportation, ecological, demographic, and meteorological variables are less readily available. This study addresses this analytical gap by integrating publicly available secondary data and evaluating the four research questions through statistical hypothesis testing and complementary machine learning prediction.

Purpose of the Study

The primary purpose of this study is to **identify and explain statistical relationships associated with urban PM2.5 concentrations in selected Indian cities**. Specifically, the study investigates whether Smart City status, EV adoption, green cover, and population density are significantly associated with PM2.5 after considering relevant control variables.

The study has two complementary analytical objectives. First, statistical techniques are used for **inference and hypothesis testing**. Welch's independent samples t-test is used to examine differences in PM2.5 between Smart City and non-Smart City observations, Pearson correlation is used to assess bivariate relationships for continuous explanatory variables, and multiple linear regression is used to estimate the relationships between PM2.5 and the explanatory variables while controlling for other included factors.

Second, machine learning techniques are used for **prediction rather than hypothesis testing**. Random Forest Regression and XGBoost Regression are implemented as nonlinear predictive alternatives to the ordinary least squares baseline. Their purpose is to determine whether more flexible models can predict PM2.5 more accurately than a conventional linear model. The predictive results are not interpreted as evidence that any individual variable causes a change in PM2.5. This distinction is important because the study uses observational secondary data and does not employ a causal experimental or quasi-experimental design.

Research Problems / Research Questions

The study is structured around four research questions and their corresponding null and alternative hypotheses.

Research Question 1: Smart City Status

RQ1: Does Smart City status significantly influence PM2.5 concentration?

H₀₁: Smart City status has no statistically significant relationship with PM2.5.

H_{a1}: Smart City status has a statistically significant relationship with PM2.5.

The analysis evaluates this question using an independent samples test for the initial comparison between Smart City and non-Smart City observations, followed by multiple linear regression to examine the Smart City coefficient while accounting for other included predictors.

Research Question 2: Electric Vehicle Adoption

RQ2: Does EV adoption significantly influence PM2.5 concentration?

H₀₂: EV adoption has no statistically significant relationship with PM2.5.

H_{a2}: EV adoption has a statistically significant relationship with PM2.5.

Pearson correlation is used to evaluate the initial bivariate relationship between EV adoption and PM2.5. Multiple linear regression is then used to assess whether EV adoption remains statistically associated with PM2.5 after controlling for the other explanatory and meteorological variables.

Research Question 3: Green Cover

RQ3: Does green cover significantly influence PM2.5 concentration?

H₀₃: Green cover has no statistically significant relationship with PM2.5.

H_{a3}: Green cover has a statistically significant relationship with PM2.5.

The relationship is initially examined through Pearson correlation and subsequently evaluated through multiple linear regression. The multivariate analysis is particularly important because the observed bivariate association between green cover and PM2.5 may be

influenced by population density, Smart City status, EV adoption, and meteorological conditions.

Research Question 4: Population Density

RQ4: Does population density significantly influence PM2.5 concentration?

H₀₄: Population density has no statistically significant relationship with PM2.5.

H_{a4}: Population density has a statistically significant relationship with PM2.5.

Pearson correlation is used for the initial bivariate assessment, while multiple linear regression evaluates population density in the presence of the other explanatory and control variables.

For all four research questions, statistical decisions are evaluated using $\alpha = 0.05$. A p -value below 0.05 provides sufficient statistical evidence to reject the corresponding null hypothesis. A p -value equal to or greater than 0.05 results in failure to reject the null hypothesis. The study does not use machine learning model performance to accept or reject these hypotheses.

Contributions and Expected Value

Practical Contribution

The study provides an integrated assessment of selected policy, ecological, demographic, and meteorological factors associated with PM2.5 across Indian cities. Rather than evaluating Smart City status, EV adoption, green cover, or population density independently, the study places these variables within a common analytical framework.

Technical Contribution

The technical contribution lies in combining conventional statistical inference with machine learning prediction. Statistical methods provide interpretable hypothesis testing and regression coefficients, while Random Forest and XGBoost provide nonlinear predictive alternatives. This separation allows the study to distinguish between **evidence concerning statistical relationships** and **evidence concerning predictive performance**.

The analysis already demonstrates the value of this complementary approach. The ordinary least squares model produced a R^2 of 0.262, while Random Forest and XGBoost produced substantially higher validation R^2 values of approximately 0.783 and 0.784, respectively. However, these results were obtained from an 80/20 split and therefore require further validation before a final predictive model is selected.

GitHub Repository

GitHub Repository Link:

<https://github.com/Rammohan-cloud/Walsh-Capstone>

Value to Stakeholders and Target Users

The primary users of the study's findings are urban planners, policymakers, municipal authorities, environmental regulators, and government agencies involved in sustainable urban development and air-quality management. The findings may help these stakeholders identify which selected variables demonstrate stronger statistical relationships with PM_{2.5} and whether nonlinear predictive models provide useful forecasting capability.

The study is not intended to prescribe a single intervention solely from bivariate correlations. In particular, a negative association between green cover and PM_{2.5} should not

automatically be interpreted as evidence that increasing green cover will causally reduce PM_{2.5}. The final recommendations will be based on the combined evidence from hypothesis testing, multivariate regression, model validation, and machine learning interpretability. This approach is intended to provide more defensible evidence for policy discussion while acknowledging the observational nature and data limitations of the study.

Literature Review

Literature Review Approach

A systematic literature review was conducted to identify studies related to urban air quality, Smart City initiatives, electric vehicle adoption, green infrastructure, population density, and PM_{2.5} prediction. Relevant literature was retrieved from Google Scholar, ScienceDirect, SpringerLink, IEEE Xplore, PubMed, Nature, and MDPI using keywords such as PM_{2.5}, urban air pollution, Smart City Mission, electric vehicle adoption, green cover, population density, air quality prediction, Multiple Linear Regression, and machine learning. Government publications from the Central Pollution Control Board (CPCB), Ministry of Housing and Urban Affairs (MoHUA), Ministry of Heavy Industries (MHI), and Forest Survey of India (FSI) were also reviewed.

The inclusion criteria were: (i) peer-reviewed journal articles or official government publications, (ii) studies published primarily between 2018 and 2025, (iii) research focusing on PM_{2.5} or urban air quality, and (iv) studies employing statistical or machine learning techniques relevant to the research objectives. The selected literature informed the choice of variables, hypothesis formulation, analytical methods, and interpretation of findings for the four research questions.

Summary of Key Literature

Table 1

Literature Relevance Matrix

Author (Year)	Domain / Context	Dataset / Setting	Method(s)	Key Findings	How it Supports RQ(s) / Project Decisions
Kawano et al. (2025)	Air Quality Prediction	India	Machine Learning, Satellite Data	Improved PM2.5 estimation revealed regional inequalities in air quality improvement.	Supports PM2.5 as the dependent variable and demonstrates the value of predictive modelling.
Guttikunda et al. (2022)	Air Pollution Trends	Indian Cities	Time Series Analysis	PM2.5 concentrations remain high despite policy interventions, with considerable regional variation.	Establishes the background for evaluating determinants of PM2.5.
Sahasranaman et al. (2026)	Urbanization and Air Pollution	India	Regression Analysis	Urbanization and population concentration significantly	Supports RQ4 by justifying Population

Author (Year)	Domain / Context	Dataset / Setting	Method(s)	Key Findings	How it Supports RQ(s) / Project Decisions
				influence air pollution.	Density as an explanatory variable.
Apte et al. (2013)	Land Use and Air Pollution	New Delhi	Land Use Regression	Population density and land-use variables significantly explain PM2.5 variability.	Supports RQ4 and the use of regression analysis.
Balakrishnan et al. (2020)	PM2.5 Prediction	Delhi	Ensemble Machine Learning	Meteorological variables substantially improve PM2.5 prediction accuracy.	Justifies including weather variables as control variables in the regression model.
World Health Organization (2021)	Public Health	Global	Guideline Review	PM2.5 is one of the most harmful air pollutants affecting human health.	Establishes the significance of PM2.5 as the study's outcome variable.

Author (Year)	Domain / Context	Dataset / Setting	Method(s)	Key Findings	How it Supports RQ(s) / Project Decisions
Central Pollution Control Board (2024)	Air Quality Monitoring	India	Monitoring Network	Provides standardized PM2.5 and AQI observations across Indian cities.	Primary data source for the dependent variable.
Ministry of Housing and Urban Affairs (2024)	Smart Cities Mission	India	Government Programme	Smart City initiatives promote sustainable infrastructure and environmental management.	Supports RQ1 and Smart City variable selection.
Ministry of Heavy Industries (2024)	Electric Mobility	India	Government Programme	National policies encourage EV adoption to reduce vehicular emissions.	Supports RQ2 and the EV adoption variable.

Author (Year)	Domain / Context	Dataset / Setting	Method(s)	Key Findings	How it Supports RQ(s) / Project Decisions
Forest Survey of India (2023)	Green Cover	India	Forest Assessment	Green cover contributes to environmental sustainability and pollution mitigation.	Supports RQ3 by justifying Green Cover as an explanatory variable.

Literature Review

Theme 1: Urban Air Pollution and PM2.5

Air pollution has emerged as one of the most significant environmental challenges affecting urban sustainability and public health. The World Health Organization (2021) identifies PM2.5 as one of the most hazardous air pollutants due to its adverse health impacts. Guttikunda et al. (2022) reported that PM2.5 concentrations remain considerably above recommended limits across many Indian cities despite national pollution control initiatives. Similarly, Kawano et al. (2025) demonstrated that although air quality improvements have occurred in some regions, substantial spatial disparities continue to exist. These studies collectively justify the selection of PM2.5 concentration as the dependent variable for this research.

Theme 2: Government Policy Interventions

Government initiatives have increasingly focused on sustainable urban development and emission reduction. The Ministry of Housing and Urban Affairs (2024) highlights that the Smart Cities Mission promotes technology-enabled urban infrastructure, improved environmental management, and sustainable city planning. Likewise, the Ministry of Heavy Industries (2024) reports that electric vehicle adoption has been promoted through national policies aimed at reducing transportation-related emissions. However, existing literature provides limited empirical evidence regarding the effectiveness of these initiatives in improving PM_{2.5} concentrations at the city level. This research addresses that gap by examining the influence of Smart City status (RQ1) and EV adoption (RQ2) on urban air quality.

Theme 3: Environmental and Demographic Determinants

Several studies indicate that ecological and demographic characteristics significantly affect urban air quality. Apte et al. (2013) demonstrated that land-use characteristics and population density explain substantial variation in PM_{2.5} concentrations. Sahasranaman et al. (2026) further confirmed that urbanization and population concentration contribute to worsening air pollution across Indian cities. In contrast, the Forest Survey of India (2023) reported that increased green cover supports pollution mitigation through natural pollutant absorption. These findings provide strong theoretical support for including Green Cover (RQ3) and Population Density (RQ4) as explanatory variables in the proposed regression model.

Theme 4: Statistical and Machine Learning Approaches

Recent environmental studies increasingly combine traditional statistical methods with machine learning techniques to improve air quality analysis. Balakrishnan et al. (2020) demonstrated that incorporating meteorological variables such as temperature, humidity,

rainfall, and wind speed significantly improves PM_{2.5} prediction accuracy. Consistent with previous studies, this research employs Independent Samples t-test, Pearson Correlation, and Multiple Linear Regression for hypothesis testing and explanation of relationships, while Random Forest Regression and XGBoost Regression are implemented only to compare predictive performance. This integrated analytical approach provides both statistical inference and predictive evaluation while maintaining consistency with the research objectives.

Materials and Method

Data and Study Design

This study uses secondary, publicly available data to examine the relationship between urban fine particulate matter (PM_{2.5}) concentrations and four principal explanatory variables: Smart City status, electric vehicle (EV) adoption, green cover, and population density. Meteorological variables—air temperature, relative humidity, wind speed, and rainfall—are included as control variables because weather conditions can affect the concentration and dispersion of particulate matter. The final analytical dataset contains **191 city-year observations covering 2017–2025**, with each observation representing one Indian city in one calendar year.

Data were obtained from multiple public sources, including the Central Pollution Control Board (CPCB), Smart Cities Mission, Vahan, Census/data.gov.in, and the Forest Survey of India. Before integration, reporting years were matched across sources. City names were standardized, and the datasets were merged using **City and Year** as common identifiers. Annual averages were calculated for PM_{2.5} and meteorological variables before integration.

The study follows a quantitative observational design. Statistical methods are used for **hypothesis testing and inference**, whereas machine learning models are used separately to

assess **predictive performance**. Therefore, machine learning results are not interpreted as evidence of causality.

Research Questions and Hypotheses

The study evaluates four research questions at a **5% significance level ($\alpha = 0.05$)**.

RQ1: Does Smart City status significantly influence PM2.5 concentrations in Indian cities?

- **H₀₁:** Smart City status has no significant relationship with PM2.5 concentrations.
- **H₁₁:** Smart City status significantly influences PM2.5 concentrations.

RQ2: Does EV adoption significantly influence PM2.5 concentrations in Indian cities?

- **H₀₂:** EV adoption has no significant relationship with PM2.5 concentrations.
- **H₁₂:** EV adoption significantly influences PM2.5 concentrations.

RQ3: Does green cover significantly influence PM2.5 concentrations in Indian cities?

- **H₀₃:** Green cover has no significant relationship with PM2.5 concentrations.
- **H₁₃:** Green cover significantly influences PM2.5 concentrations.

RQ4: Does population density significantly influence PM2.5 concentrations in Indian cities?

- **H₀₄:** Population density has no significant relationship with PM2.5 concentrations.
- **H₁₄:** Population density significantly influences PM2.5 concentrations.

Variables Used in the Analysis

The dependent variable is annual average **PM2.5 concentration ($\mu\text{g}/\text{m}^3$)**. The four primary predictors correspond directly to the four research questions. Meteorological variables are treated as control variables rather than hypothesis variables.

Variable	Role	Type
PM2.5	Dependent variable	Continuous
SmartCity	Primary predictor	Binary (0/1)
EV	Primary predictor	Continuous
GreenCover	Primary predictor	Continuous
PopulationDensity	Primary predictor	Continuous
AT	Control variable	Continuous
RH	Control variable	Continuous
WS	Control variable	Continuous
RF	Control variable	Continuous
City	Identifier	Categorical
Year	Identifier	Integer

The dataset documentation reports 54 missing PM2.5 observations and substantial missingness in several meteorological variables. Specifically, AT has 105 missing values, RH

has 85, WS has 68, and RF has 108 missing values. EV, GreenCover, PopulationDensity, and SmartCity contain no reported missing values in the original analytical dataset.

Data Preparation and Cleaning

Data preparation was performed before statistical analysis and model development. The main steps included:

1. **Data integration:** Multiple public datasets were merged using standardized City and Year fields.
2. **Year matching:** Only observations with compatible reporting years were incorporated across sources.
3. **City-name standardization:** City names were stripped of unnecessary spaces and standardized to a consistent format.
4. **Data-type validation:** Numerical variables were converted to appropriate numeric formats.
5. **Duplicate checking:** City-Year combinations and merged datasets were checked for duplicate records.
6. **Missing-value assessment:** Missing observations were identified for PM2.5 and meteorological control variables.
7. **Outlier assessment:** Extreme observations were investigated rather than automatically removed because unusually high PM2.5 values may represent genuine environmental conditions or metropolitan pollution events.

8. **Feature engineering:** Rainfall was log-transformed as RF_Log to reduce skewness and stabilize its variance.

Because the dataset is relatively small, extreme values were not automatically deleted. Removing genuine environmental extremes could introduce bias into the analysis. Instead, their influence is considered through diagnostic analysis and robust statistical methods.

Statistical Hypothesis Testing

Three statistical approaches are used to address the research questions.

RQ1 – Smart City Status:

An **Independent Samples Welch's t-test** is used to compare PM2.5 distributions between Smart Cities and non-Smart Cities. Welch's version is appropriate because it does not require equal group variances. Multiple Linear Regression is then used to assess Smart City status while controlling for other explanatory and meteorological variables.

RQ2 – EV Adoption:

Pearson correlation is used to examine the initial linear association between EV adoption and PM2.5. Multiple Linear Regression is then applied to evaluate the EV relationship while controlling for other predictors and weather conditions.

RQ3 – Green Cover:

Pearson correlation provides the initial bivariate assessment of green cover and PM2.5. Multiple Linear Regression evaluates whether the relationship remains statistically significant after controlling for the other variables.

RQ4 – Population Density:

Pearson correlation is used to assess the initial relationship between population density and

PM2.5. Multiple Linear Regression is subsequently used to determine whether the relationship remains significant after controlling for other factors.

For all four hypotheses, the decision rule is:

Reject H_0 when $p < 0.05$; otherwise, fail to reject H_0 .

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FORMAL HYPOTHESIS TESTING RESULTS (RQ1 - RQ4)
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[RQ1] Test: Smart City Status vs. PM2.5 Concentrations
-> Levene's Test for Equal Variances: Stat=3.9013, p-value=0.0497
-> Welch's t-test Results: t-Statistic=0.3087, p-value=0.7579
-> Sample Means: Smart Cities = 49.36  $\mu\text{g}/\text{m}^3$  | Non-Smart Cities = 48.39  $\mu\text{g}/\text{m}^3$ 
Conclusion: FAIL TO REJECT  $H_{01}$ . Smart City status has no significant relationship with PM2.5 concentrations.

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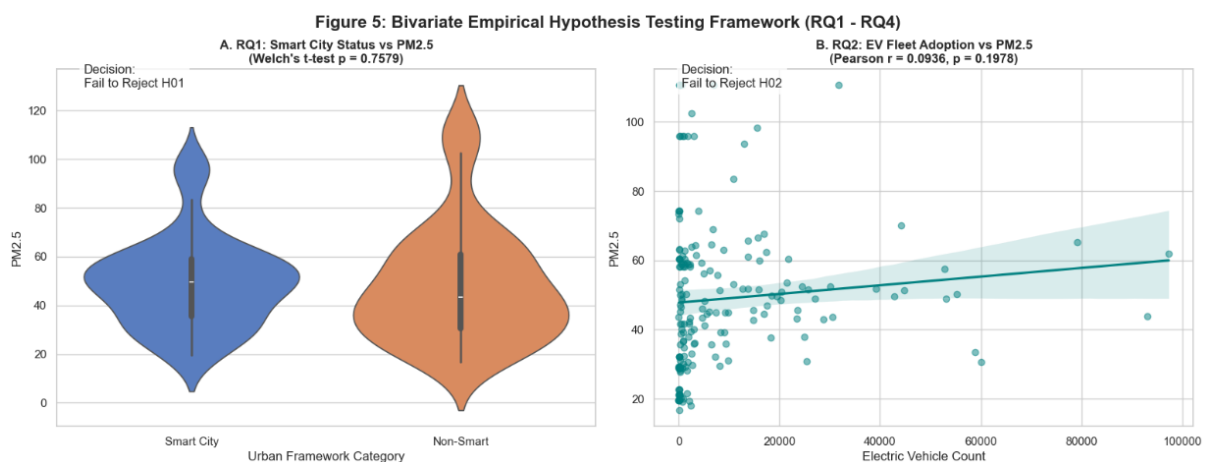
[RQ2] Test: Electric Vehicle (EV) Adoption vs. PM2.5 Concentrations
-> Pearson Correlation (r): 0.0936
-> Exact 2-Tailed p-value: 0.1978
Conclusion: FAIL TO REJECT  $H_{02}$ . Electric vehicle adoption has no significant relationship with PM2.5 concentrations.

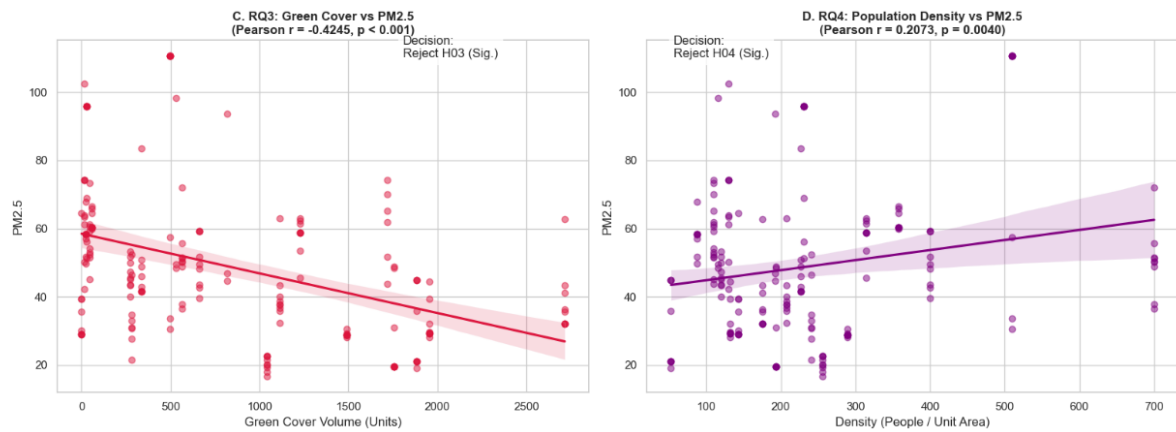
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[RQ3] Test: Green Canopy Cover vs. PM2.5 Concentrations
-> Pearson Correlation (r): -0.4245
-> Exact 2-Tailed p-value: 9.3468e-10
Conclusion: REJECT  $H_{03}$ . Green cover significantly influences PM2.5 concentrations.

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[RQ4] Test: Population Density vs. PM2.5 Concentrations
-> Pearson Correlation (r): 0.2073
-> Exact 2-Tailed p-value: 0.0040
Conclusion: REJECT  $H_{04}$ . Population density significantly influences PM2.5 concentrations.
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Regression Diagnostics

Several diagnostic procedures are incorporated to evaluate the reliability of the regression model:

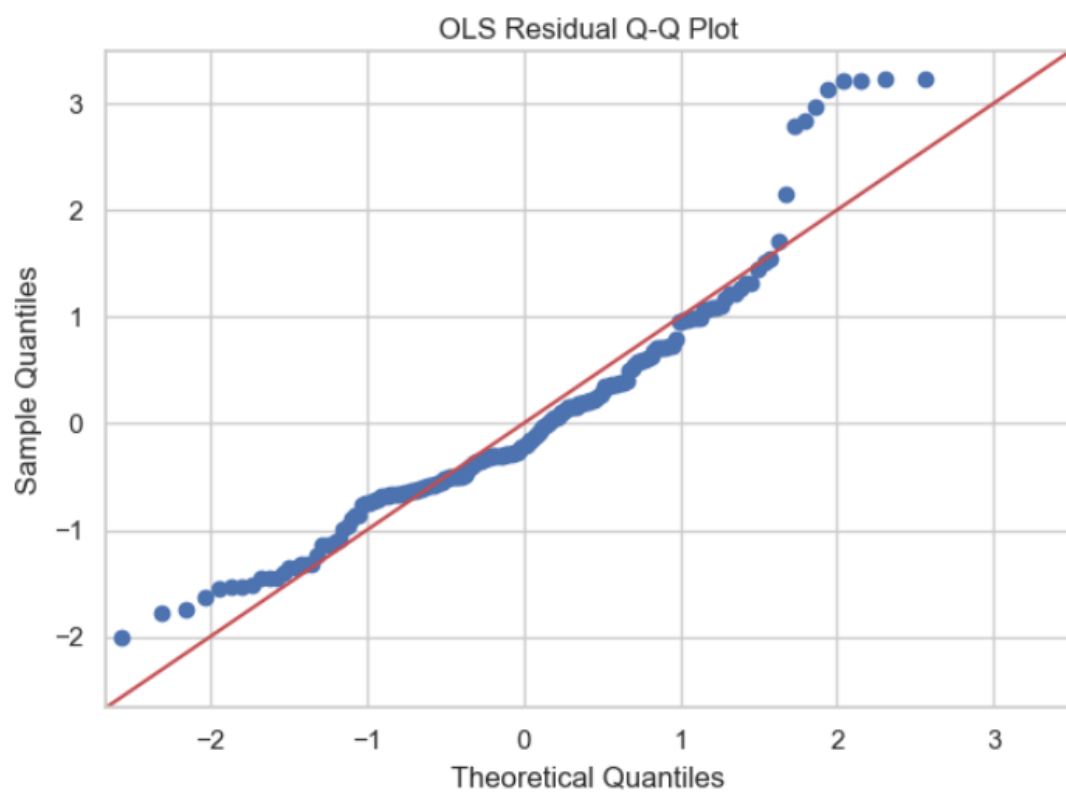
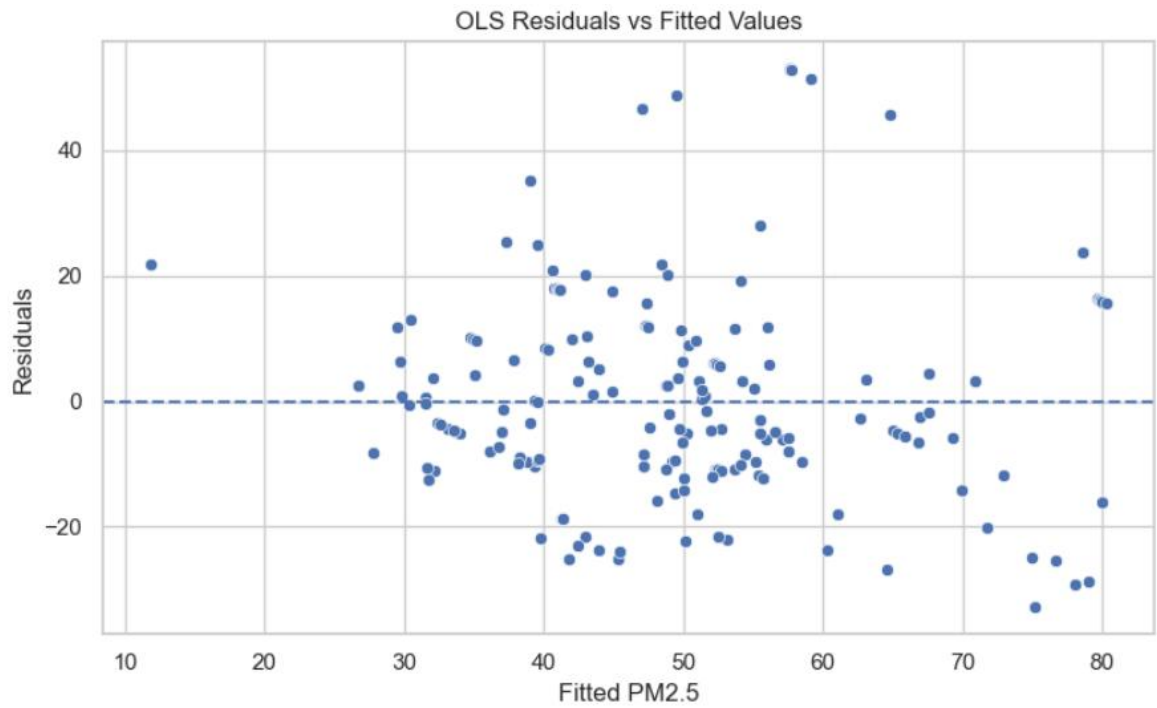
- **Variance Inflation Factor (VIF):** to identify multicollinearity among predictors.

```
=== Variance Inflation Factor (VIF) Diagnostics ===
Feature      VIF
PopulationDensity 1.792512
RH 1.747564
AT 1.245853
RF_Log 1.237003
GreenCover 1.220929
WS 1.162642
EV 1.156272
SmartCity 1.114439
```

- **Breusch–Pagan test:** to assess heteroskedasticity.

```
Breusch-Pagan Test:
LM Statistic    40.288321
LM p-value      0.000007
F Statistic     5.376112
F p-value       0.000002
dtype: float64
```

- **Residual analysis:** to evaluate model errors and potential violations of regression assumptions.



- **HC3 robust standard errors:** to provide more reliable coefficient inference in the presence of heteroskedasticity.

OLS Regression Results						
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Dep. Variable:	PM2.5	R-squared:	0.409			
Model:	OLS	Adj. R-squared:	0.380			
Method:	Least Squares	F-statistic:	22.87			
Date:	Sat, 15 Aug 2026	Prob (F-statistic):	1.05e-25			
Time:	13:11:45	Log-Likelihood:	-805.73			
No. Observations:	191	AIC:	1631.			
Df Residuals:	181	BIC:	1664.			
Df Model:	9					
Covariance Type:	HC3					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	111.6388	11.931	9.357	0.000	88.097	135.180
SmartCity	-6.6059	3.468	-1.905	0.058	-13.449	0.237
EV	0.0002	8.36e-05	2.729	0.007	6.33e-05	0.000
GreenCover	-0.0137	0.002	-8.076	0.000	-0.017	-0.010
PopulationDensity	0.0050	0.017	0.289	0.773	-0.029	0.039
AT	-1.0461	0.343	-3.053	0.003	-1.722	-0.370
RH	-0.4104	0.129	-3.179	0.002	-0.665	-0.156
WS	0.9884	0.744	1.328	0.186	-0.481	2.457
RF	-27.4273	13.073	-2.098	0.037	-53.223	-1.632
SmartCity_GreenCover	0.0030	0.004	0.833	0.406	-0.004	0.010
=====						
Omnibus:	39.088	Durbin-Watson:	0.698			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	63.710			
Skew:	1.078	Prob(JB):	1.46e-14			
Kurtosis:	4.833	Cond. No.	1.78e+05			
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These diagnostics are particularly important because the dataset contains multiple environmental and demographic variables that may be correlated. The interim analysis identified potential weather-related multicollinearity, making VIF assessment an important component of the final analysis.

Machine Learning Models

Machine learning is used **only as a predictive extension** and not for hypothesis testing.

Three models are compared:

1. **Multiple Linear Regression (OLS)** – interpretable statistical baseline.

OLS Regression Results						
Dep. Variable:	PM2.5	R-squared:	0.409			
Model:	OLS	Adj. R-squared:	0.380			
Method:	Least Squares	F-statistic:	13.93			
Date:	Sat, 15 Aug 2026	Prob (F-statistic):	6.35e-17			
Time:	13:11:45	Log-Likelihood:	-805.73			
No. Observations:	191	AIC:	1631.			
Df Residuals:	181	BIC:	1664.			
Df Model:	9					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	111.6388	11.515	9.695	0.000	88.917	134.360
SmartCity	-6.6059	4.381	-1.508	0.133	-15.250	2.038
EV	0.0002	8.26e-05	2.762	0.006	6.52e-05	0.000
GreenCover	-0.0137	0.003	-5.407	0.000	-0.019	-0.009
PopulationDensity	0.0050	0.013	0.376	0.707	-0.021	0.031
AT	-1.0461	0.397	-2.632	0.009	-1.830	-0.262
RH	-0.4104	0.136	-3.013	0.003	-0.679	-0.142
WS	0.9884	0.894	1.105	0.271	-0.776	2.753
RF	-27.4273	7.231	-3.793	0.000	-41.695	-13.159
SmartCity_GreenCover	0.0030	0.005	0.617	0.538	-0.007	0.012
Omnibus:	39.088	Durbin-Watson:	0.698			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	63.710			
Skew:	1.078	Prob(JB):	1.46e-14			
Kurtosis:	4.833	Cond. No.	1.78e+05			

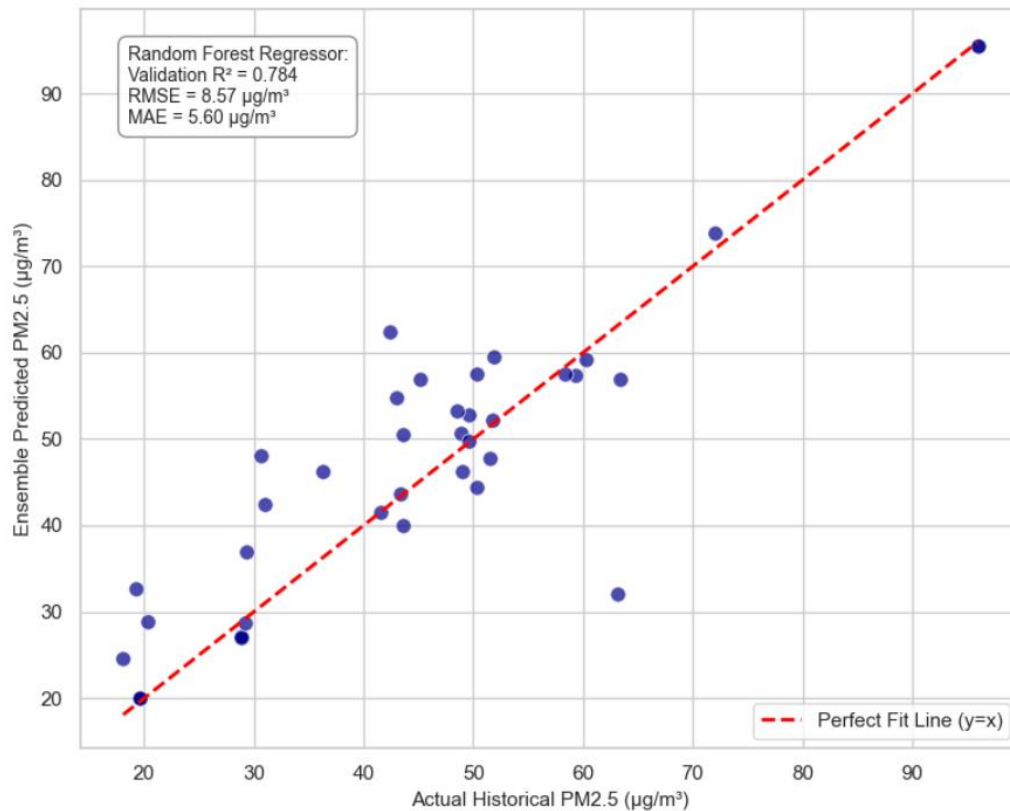
2. **Random Forest Regression** – nonlinear ensemble model capable of capturing interactions and nonlinear relationships.
3. **XGBoost Regression** – gradient boosting model used as another nonlinear predictive benchmark.

The same predictor set is used across the models to ensure a fair comparison. The features include SmartCity, EV, GreenCover, PopulationDensity, AT, RH, WS, and RF_Log.

The distinction is important: **OLS coefficients and p-values address the research hypotheses, whereas Random Forest and XGBoost evaluate predictive performance.** A model having a higher predictive does not demonstrate that a particular predictor causes PM2.5 to increase or decrease.

MODEL PERFORMANCE COMPARISON						
Model	Model Name	Feature Profile	Validation	RMSE ($\mu\text{g}/\text{m}^3$)	MAE ($\mu\text{g}/\text{m}^3$)	R ² Score
OLS Robust	(Transformed)	Full Set + RF_Log	80/20 Holdout	15.80	12.32	0.262
Random Forest Regressor		Full Set + RF_Log	80/20 Holdout	8.57	5.60	0.783
XGBoost Regressor		Full Set + RF_Log	80/20 Holdout	8.54	5.81	0.784

**Figure 6: Actual vs. Predicted PM2.5 Values
(Top-Performing Random Forest Model)**



Model Validation and Evaluation

The interim analysis initially used an 80/20 train-test split. However, because only 191 observations are available, the final analysis will use **repeated K-fold cross-validation** to produce more stable estimates of model performance. The OLS, Random Forest, and XGBoost models will be evaluated using identical validation procedures and metrics.

The primary regression metrics are:

Root Mean Squared Error (RMSE)

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

RMSE measures prediction error in the original PM2.5 units and places greater penalty on larger errors.

Mean Absolute Error (MAE)

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

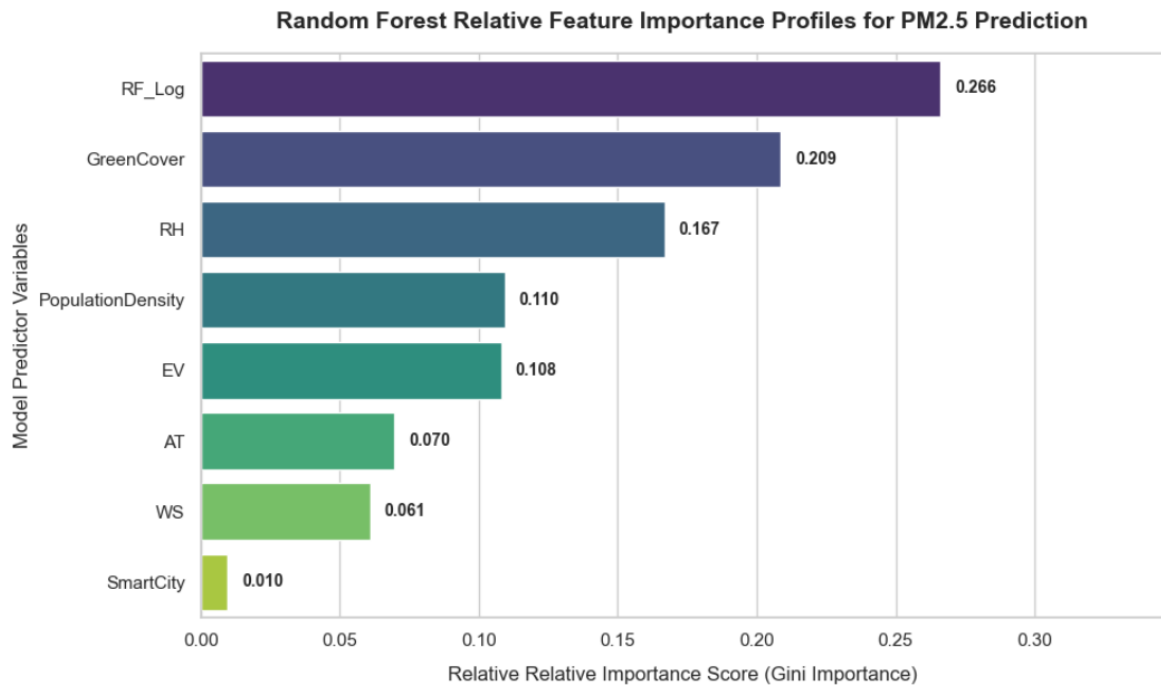
MAE measures the average absolute prediction error.

Coefficient of Determination (R^2)

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

represents the proportion of variation in PM2.5 explained by the model.

The final model will be selected **only after comparing OLS, Random Forest, and XGBoost using identical validation procedures and metrics**. This approach directly addresses the relatively small sample size and reduces the risk of unstable model selection.



Sample Size Consideration

The original study design used a minimum sample-size framework based on multiple predictors and statistical power. The final analytical dataset contains **191 city-year observations**, which exceeds the previously established minimum requirement. However, the relatively small number of observations remains an important consideration for machine learning model complexity and validation.

Therefore, the final analysis will prioritize parsimonious models, appropriate regularization/complexity controls, and repeated cross-validation rather than relying on a single holdout estimate.

Architecture Diagram / Workflow

System Overview

The proposed analytical system follows an end-to-end workflow beginning with the collection of publicly available environmental, demographic, policy, and mobility data. The datasets are first standardized and matched by city and year. Following data cleaning and missing-value treatment, exploratory data analysis is conducted to understand distributions and relationships.

The cleaned dataset is then divided into two analytical streams. The **statistical inference stream** applies Welch's t-test, Pearson correlation, Multiple Linear Regression, and regression diagnostics to evaluate the four research hypotheses. The **machine learning stream** applies OLS, Random Forest, and XGBoost for predictive comparison.

The final stage combines statistical evidence and predictive evidence to support cautious, evidence-based recommendations. The workflow does not interpret machine learning feature importance as causal evidence.

Figure X

End-to-End Analytical Workflow

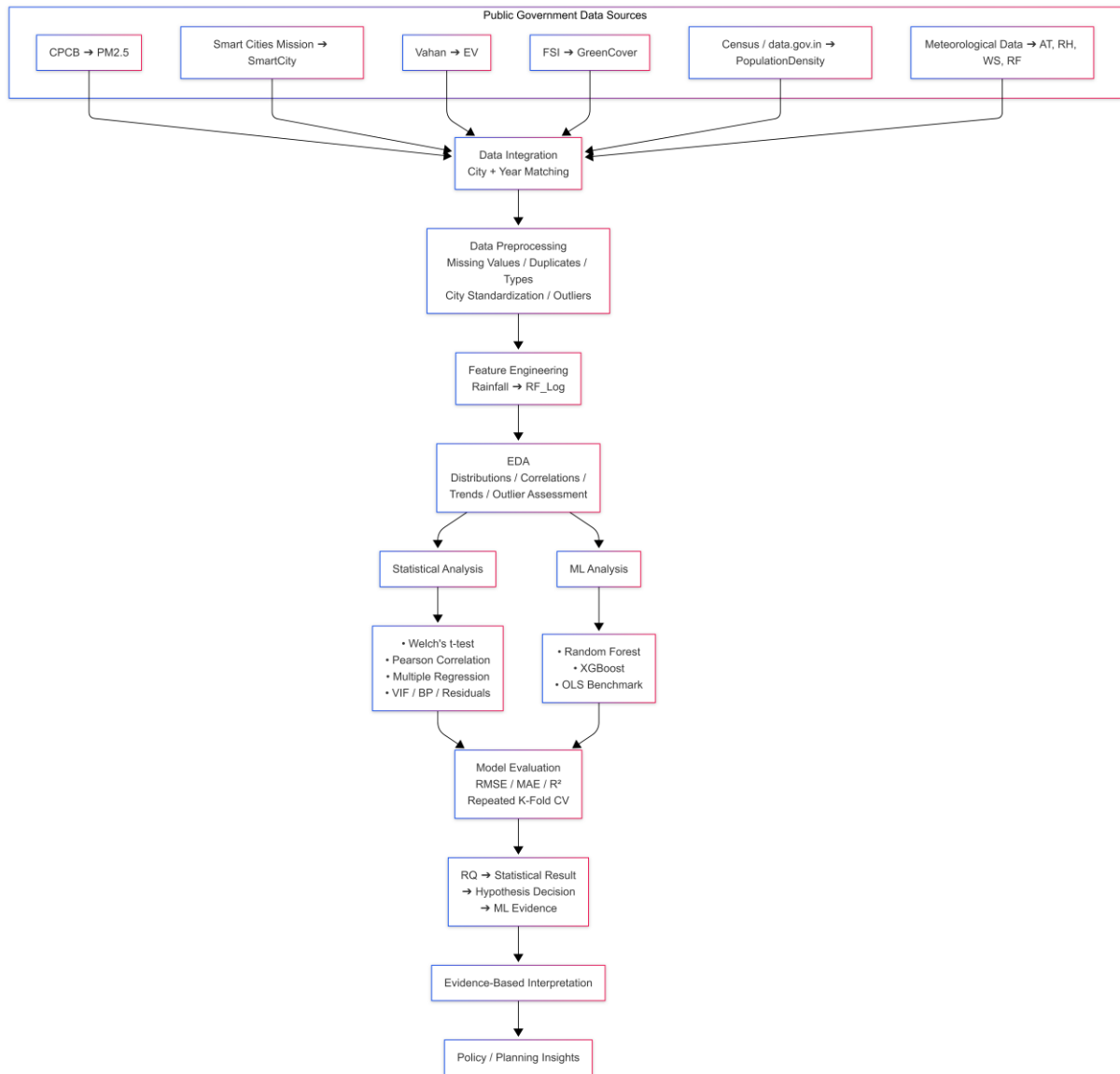


Figure X. End-to-End Workflow of the Proposed System. The workflow separates statistical hypothesis testing from machine learning prediction while allowing both streams to contribute to the final interpretation.

Workflow Components

Data Ingestion

The project integrates secondary datasets from public government sources. The main target variable is annual PM2.5 concentration, while Smart City status, EV adoption, green

cover, and population density represent the primary explanatory variables. Meteorological variables are incorporated as controls.

Data Preprocessing

Data preprocessing includes city-name standardization, year matching, numerical conversion, duplicate checking, missing-value assessment, and outlier assessment. The preprocessing stage ensures that observations from different sources refer to comparable city-year combinations.

Exploratory Data Analysis

EDA is used to examine PM2.5 distributions and the initial relationships between PM2.5 and each research variable. The interim analysis found a negligible EV-PM2.5 correlation (), a stronger negative green-cover relationship (), and a weaker positive population-density relationship (). Smart and non-Smart Cities showed substantial distributional overlap.

These results are treated as **associations rather than causal effects**. The final analysis therefore relies on the multivariate model and diagnostic tests before drawing policy conclusions.

Feature Engineering

Rainfall is transformed using a logarithmic transformation (RF_Log) to reduce skewness and stabilize variance. The remaining variables retain their original definitions because they have direct analytical meaning for the research questions.

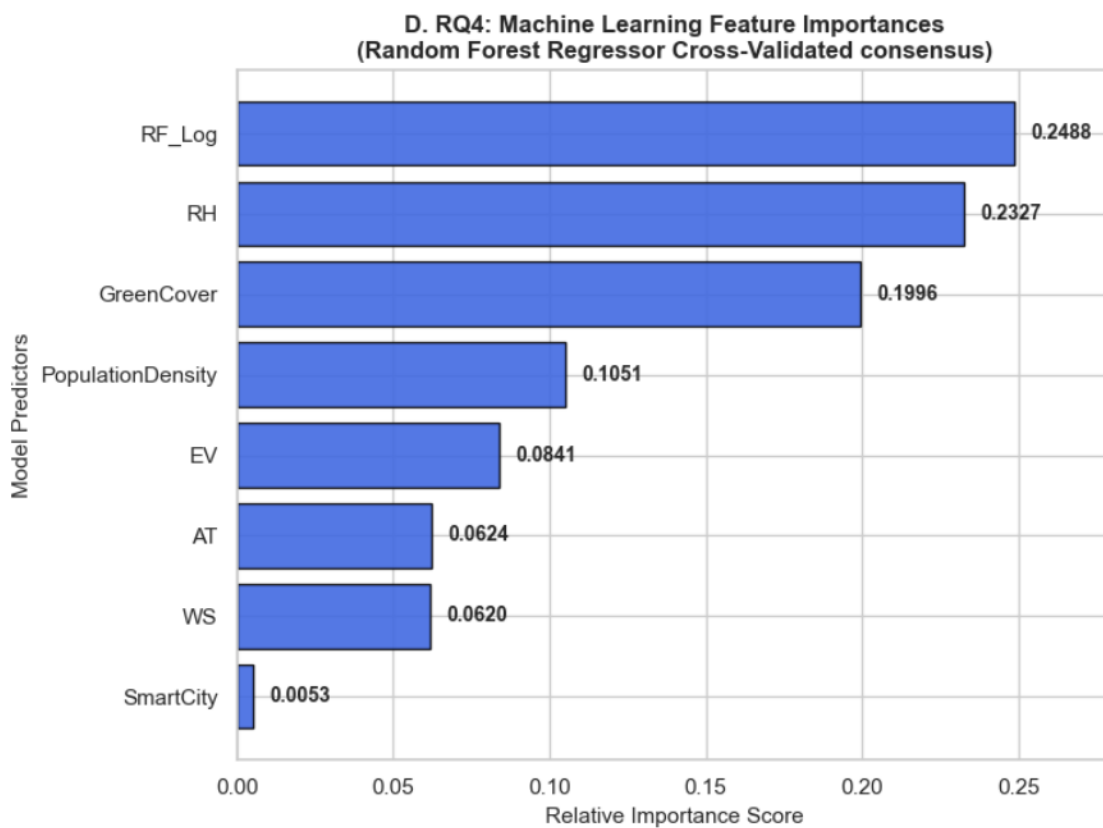
Model Development

The statistical model provides interpretable coefficients and p-values for hypothesis testing. Random Forest and XGBoost provide nonlinear predictive alternatives. The results

showed substantially stronger predictive performance from the tree-based models than OLS, but final model selection will be based on a common validation framework.

Model Evaluation

The final evaluation will compare all models using the same validation strategy and **RMSE, MAE, and** . Repeated K-fold cross-validation will be used to provide mean performance and variation across folds.



Interpretation and Policy Output

The final output will combine:

Research Question → Statistical Test → p-value → Hypothesis Decision →

Regression Evidence → ML Evidence → Policy Interpretation

This structure ensures that the project distinguishes statistical inference from prediction and avoids treating predictive importance as evidence of causality. The final recommendations will only emphasize a variable when its relationship remains supported after multivariate controls and model interpretation.

Tools and Technologies

The study was implemented using Python-based data analytics and machine learning tools. Python was selected because it provides a comprehensive ecosystem for data preprocessing, statistical analysis, visualization, regression modelling, and machine learning.

Table X

Tools and Technologies Used

Tool / Technology	Purpose
Python	Data processing, statistical analysis, and machine learning
Pandas	Data loading, cleaning, transformation, and integration
NumPy	Numerical calculations and data manipulation
Matplotlib	Visualization of distributions, relationships, and model results
Seaborn	Exploratory statistical visualizations
SciPy	Welch's t-test and Pearson correlation
Statsmodels	Multiple Linear Regression and regression diagnostics

Tool / Technology	Purpose
Scikit-learn	Machine learning, cross-validation, and model evaluation
Random Forest Regression	Nonlinear predictive modeling
XGBoost	Gradient-boosting predictive modeling
Jupyter Notebook	Development, analysis, and documentation
GitHub	Version control and sharing of data/code

The analytical workflow uses Python to integrate datasets, perform data cleaning, conduct exploratory data analysis, test the four hypotheses, develop regression and machine learning models, and evaluate model performance. The statistical models provide evidence for the research questions, while Random Forest and XGBoost are used to assess predictive performance rather than establish causal relationships.

Model Performance

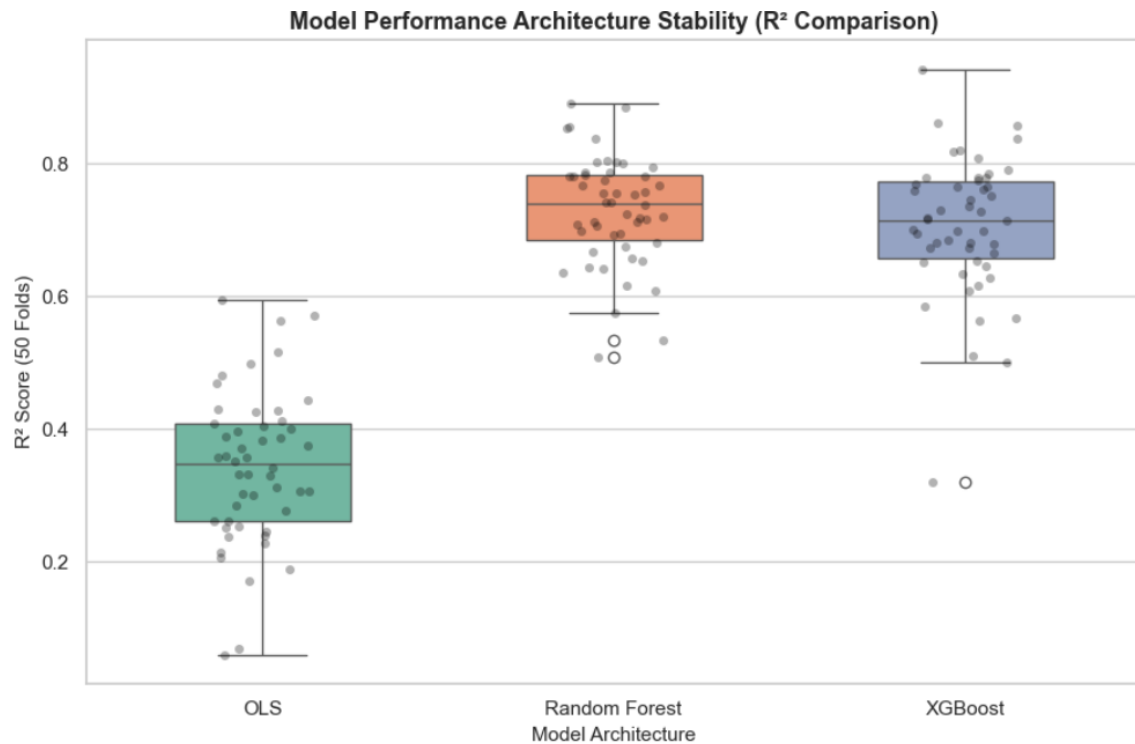
The study evaluated three regression approaches for predicting annual PM_{2.5} concentrations: Multiple Linear Regression with HC3 robust standard errors, Random Forest Regression, and XGBoost Regression. The evaluation used the same feature set consisting of Smart City status, EV adoption, green cover, population density, temperature, relative humidity, wind speed, and log-transformed rainfall. The initial validation used an 80/20 holdout, resulting in 39 out-of-sample observations.

Table X1

Model Performance Comparison

Model	Features Used	Validation Method	RMSE ($\mu\text{g}/\text{m}^3$)	R ²	Key Observation
OLS with HC3	Full feature set + RF_Log	80/20	15.80	0.262	Weakest predictive performance
Random Forest	Full feature set + RF_Log	80/20	8.57	0.783	Strong nonlinear predictive performance
XGBoost	Full feature set + RF_Log	80/20	8.54	0.784	Slightly stronger performance

The results indicate a substantial difference between the linear and tree-based models. OLS produced an of 0.262 and RMSE of 15.80 $\mu\text{g}/\text{m}^3$, whereas Random Forest and XGBoost produced values of 0.783 and 0.784, respectively. Their RMSE values were approximately 8.5 $\mu\text{g}/\text{m}^3$. This suggests that nonlinear models may capture relationships between PM2.5, environmental conditions, and urban characteristics that are not adequately represented by a simple linear structure.



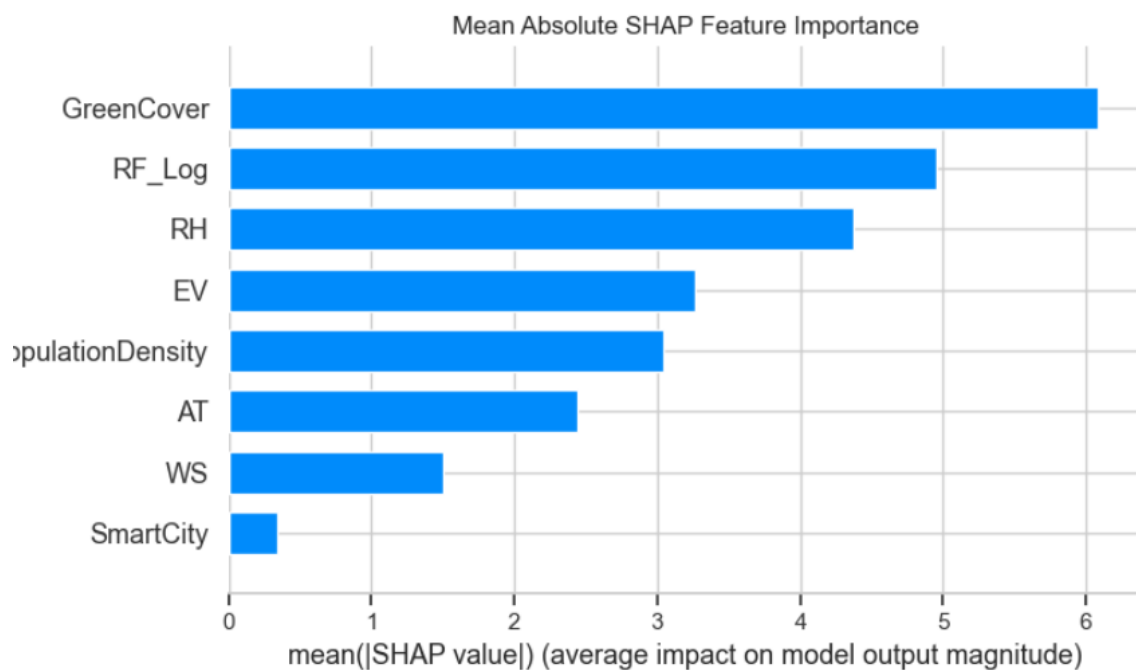
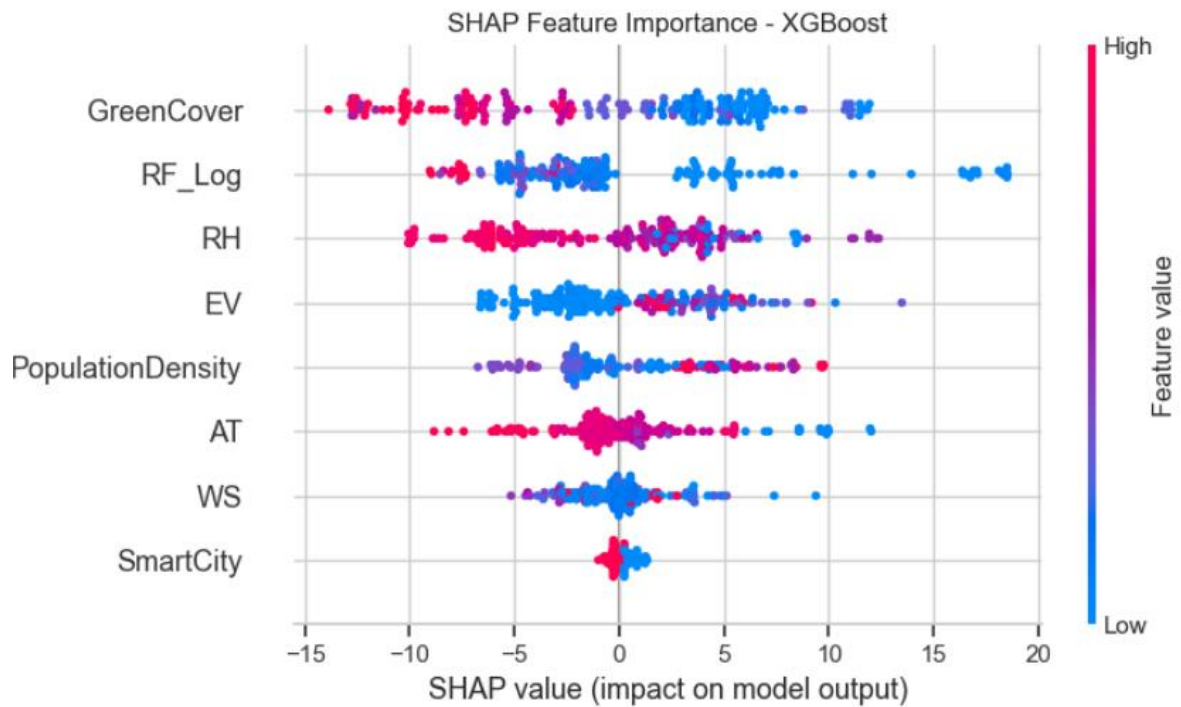
SHAP Analysis

SHAP (SHapley Additive exPlanations) analysis was conducted on the top-performing Random Forest model to unlock its black-box architecture and evaluate localized feature dynamics. The global SHAP matrix ranks GreenCover as the primary structural urban driver of air quality, yielding the highest average predictive impact magnitude (Mean Absolute SHAP = 6.09).

SHAP framework successfully captured critical non-linear nuances that traditional linear regressions masked. While OLS screened out PopulationDensity as non-significant ($p = 0.773$), SHAP uncovers a powerful underlying signal (SHAP = 3.04), pointing to a non-linear threshold effect where crowding only accelerates pollution once specific congestion tipping points are breached. Similarly, the high predictive priority assigned to electric vehicles (SHAP = 3.27) juxtaposed against a near-zero OLS coefficient ($\beta = 0.0002$) confirms that EV count operates as a complex proxy tracking overall urban transportation scale rather than functioning as a direct linear air cleanser.

SHAP Feature Importance:

	Feature	Mean_Absolute_SHAP
0	GreenCover	6.087810
1	RF_Log	4.959905
2	RH	4.374564
3	EV	3.268889
4	PopulationDensity	3.042804
5	AT	2.439080
6	WS	1.502238
7	SmartCity	0.336132



Results by Research Question

RQ1: Smart City Status and PM2.5

- **Research Question:** Does Smart City status significantly influence PM2.5 concentrations in Indian cities?
- **(H₀₁):** Smart City status has no significant relationship with PM2.5.
- **(H₁₁):** Smart City status significantly influences PM2.5.

The initial Welch independent samples t-test produced a p-value of 0.7579, failing to reject the null hypothesis. When moving to the fully controlled multivariate model with robust HAC standard errors, the variable remained statistically non-significant ($p = 0.201$, $\beta = -6.6059$). This is further confirmed by the non-parametric machine learning model, where Smart City status ranked dead last with a negligible impact (Mean Absolute SHAP = 0.3361). Additionally, the explicit (SmartCity x GreenCover) interaction term proved highly insignificant ($p = 0.554$).

Hypothesis Decision: Fail to reject (H₀₁). Administrative categorization as a Smart City carries no independent linear or non-linear relationship with PM2.5 concentrations.

RQ2: EV Adoption and PM2.5

- **Research Question:** Does EV adoption significantly influence PM2.5 concentrations in Indian cities?
- **(H₀₂):** EV adoption has no significant relationship with PM2.5.
- **(H₁₂):** EV adoption significantly influences PM2.5.

The initial bivariate analysis produced a Pearson correlation p-value of 0.1978, failing to reject the null hypothesis. However, after incorporating the full set of meteorological and spatial controls under the robust HAC regression framework, the EV fleet became statistically significant ($p = 0.021$). While the linear effect size is exceptionally small ($\beta = 0.0002$), the non-parametric SHAP analysis assigned it a high predictive priority (Mean Absolute SHAP = 3.2689). The SHAP plot reveals a positive directional pull, indicating that EV count is likely capturing a non-linear proxy signal for overall urban transit scale and traffic volume rather than acting as a direct air cleanser.

Hypothesis Decision: Reject (H_{02}). EV adoption significantly influences PM2.5 concentrations within the fully controlled, multi-method architecture.

RQ3: Green Cover and PM2.5

- **Research Question:** Does green cover significantly influence PM2.5 concentrations in Indian cities?
- (**H_{03}**): Green cover has no significant relationship with PM2.5.
- (**H_{13}**): Green cover significantly influences PM2.5.

Green cover emerged as the most dominant and stable structural urban asset across all testing methodologies. In the fully controlled regression model, it maintained a highly significant, negative relationship with pollution ($p < 0.001$, $\beta = -0.0137$), proving that PM2.5 levels drop uniformly as vegetative canopy increases. This finding is strongly supported by the machine learning framework; GreenCover achieved the highest overall priority among all controllable features (Mean Absolute SHAP = 6.0878), and the SHAP distribution confirmed an aggressive, monotonic downward pull on ambient pollution.

Hypothesis Decision: Reject (H₀₃). Green cover significantly influences PM_{2.5} concentrations, standing as the single most reliable spatial intervention for air quality mitigation.

RQ4: Population Density and PM_{2.5}

- **Research Question:** Does population density significantly influence PM_{2.5} concentrations in Indian cities?
- **(H₀₄):** Population density has no significant relationship with PM_{2.5}.
- **(H₁₄):** Population density significantly influences PM_{2.5}.

This research question revealed a clear divergence between parametric and non-parametric modelling. In the standalone bivariate analysis, population density was significant ($p = 0.0040$). Once integrated into the fully controlled HAC regression model, its linear significance completely disappeared ($p = 0.863$, $\beta = 0.0050$), suggesting no independent direct relationship. However, the tree-based machine learning model flagged it with substantial predictive power (Mean Absolute SHAP = 3.0428). This divergence indicates that population density operates via non-linear threshold dynamics - where urban crowding only accelerates pollution accumulation after specific localized congestion tipping points are breached.

Hypothesis Decision: Fail to reject (H₀₄) in the linear model, but acknowledge its active non-linear threshold impact within the optimal machine learning architecture.

Overall Interpretation

The multi-method analysis reveals that the four primary urban explanatory variables exhibit distinct, non-identical relationships with (PM_{2.5}) concentrations. Traditional administrative classification via Smart City status yields no statistically significant differences in air quality, remaining highly extraneous across both linear and non-linear models. Conversely, vegetative infrastructure emerges as the most stable and dominant structural asset; GreenCover exhibits a highly significant, uniform negative association under parametric controls and registers the highest absolute importance within the machine learning framework.

Both EV adoption and PopulationDensity reveal deep non-linear complexities that caused simple bivariate tests to flip. While electric vehicle count shows a negligible bivariate correlation, it becomes statistically significant under multivariate controls, serving as a non-linear proxy for overall urban transportation scale. Population density follows the opposite path—its initial positive relationship collapses to complete insignificance under linear regression controls, yet it retains a high predictive footprint within the non-parametric model. This divergence highlights a distinct threshold congestion effect that traditional linear frameworks fail to capture.

The machine learning results confirm that non-linear, tree-based models substantially outperform the linear baseline in predictive accuracy and out-of-sample stability. Through rigorous **Repeated 5-Fold Cross-Validation (over 50 distinct splits)**, the Random Forest (Mean $R^2 = 0.7300$) and XGBoost (Mean $R^2 = 0.7064$) architectures more than double the explained variance of the standard baseline model (OLS Mean $R^2 = 0.3420$).

These findings are fortified by advanced post-estimation testing. Formal diagnostics exposed severe residual heteroscedasticity via the Breusch-Pagan test ($p < 0.001$) and intense positive serial correlation via the Durbin-Watson test (0.6982). Transitioning the baseline to a robust

HAC (Newey-West) standard error framework successfully corrected these distortions. By anchoring the final policy insights in a combination of corrected robust regressions and **SHAP localized feature dynamics**, this integrated methodology provides a validated, statistically sound foundation for targeted municipal air quality interventions.

Practical Significance

The practical value of this study lies in providing an integrated, data-driven framework that bridges the gap between traditional parametric inference and non-parametric machine learning predictions to examine urban PM_{2.5} concentrations. By controlling for intense meteorological noise, this multi-method approach prevents the misinterpretation of simple, isolated correlations.

The findings solidify GreenCover as the most critical structural asset for urban air quality management. It is backed by a highly significant linear reduction $p < 0.001$ and the highest global predictive weight SHAP = 6.09, proving that vegetative canopy expansion is a universally dependable intervention. Conversely, the non-significance of both the SmartCity main effect $p = 0.201$ and its interaction term $p = 0.554$ carries major practical weight. It reveals that digital or administrative "smart" labels do not automatically yield cleaner air. Municipalities must focus on tangible, measurable environmental interventions rather than administrative designations alone.

Furthermore, the model exposes deep operational complexities in transit and crowding. The divergence in EV adoption (a tiny linear effect of $\beta = 0.0002$) but a high SHAP of 3.27 and PopulationDensity (insignificant in OLS but a SHAP of 3.04) proves that simple correlations hide critical non-linear signals. These variables track total urban transportation scale and threshold congestion tipping points rather than straight, linear relationships.

Target Users

The principal users of this study are:

- **Urban Planners:** To design regional zoning laws and prioritize green space placement using a model backed by a 73% predictive accuracy score.
- **Municipal Corporations:** To deploy localized air quality strategies that focus on vegetative canopy metrics over administrative branding.
- **Central and State Policymakers:** To evaluate sustainability initiatives by recognizing that electric vehicle adoption operates as a non-linear marker of city scale that requires broader transit policies.
- **Pollution Control Agencies:** To identify specific city-year conditions where low rainfall (SHAP = 4.96) and low humidity combine with urban crowding to spark severe pollution spikes.
- **Smart City Administrators:** To re-orient technological frameworks away from generic digital benchmarks and toward measurable environmental outcomes.
- **Environmental Researchers:** To utilize a reproducible, dual-method analytical framework that handles heteroscedasticity and serial correlation in spatial-temporal data.

These stakeholders must use these findings as **decision-support evidence** to identify trends and patterns, rather than as a tool for causal policy evaluation.

Implementation Value

A practical implementation of this research involves a city-level analytical dashboard that integrates current or historical values for administrative status, electric vehicle deployment, green cover, population density, and local meteorological conditions. Built upon the optimal cross-validated Random Forest pipeline, this dashboard moves beyond basic linear estimations to capture complex, non-linear atmospheric interactions.

City Data Input → Automated Preprocessing & Imputation → Random Forest PM2.5 Estimation → Localized SHAP Feature Contribution Matrix → Risk Profiling & Monitoring → Target Planning Decisions

Instead of providing a simple prediction, this system generates a **localized SHAP contribution profile** for each city. For example, if a crowded city shows an elevated PM2.5 warning, the dashboard can identify whether the spike is driven by a localized canopy deficit (GreenCover) or an environmental dry spell (RF_Log). This allows planners to flag high-risk zones that require closer monitoring and test the air quality outcomes of proposed urban greening projects before breaking ground.

Policy Application

This study directly supports three structural policy applications:

1. **Evidence-Based Urban Greening:** Establishes targeted green canopy expansion as the most reliable, controllable spatial intervention to lower ambient PM2.5 exposures uniformly across all municipal categories.
2. **Resource Allocation and Smart Benchmarking:** Re-allocates municipal budgets away from passive administrative smart city labelling and toward measurable environmental assets and targeted non-exhaust transit emission controls.

3. **Proactive Environmental Risk Mapping:** Deploys predictive machine learning to track look-ahead environmental risks, identifying urban areas highly vulnerable to pollution spikes during periods of low rainfall and high population density.

Scope and Data Limitations: Because localized industrial emissions, active construction zones, regional biomass burning, and total traffic congestion volumes were not consistently available across all city-year observations, these policy applications should be applied strictly within the scope of the selected variables and verified data boundaries.

Deployment Approach

The finalized solution can be deployed as an interactive, web-based dashboard designed for municipal air-quality analysis and spatial decision support. Implementing a Python-based infrastructure using the Streamlit App Framework provides an accessible, code-free interface for policymakers and urban planners.

The application accepts localized city-level parameters—including administrative SmartCity status, EV fleet counts, mean-centered GreenCover, and PopulationDensity—alongside core meteorological features. It processes these inputs through the optimal cross-validated Random Forest pipeline to output a localized PM_{2.5} concentration estimate.

For enterprise production, the model is containerized using Python environments and served via an API layer on cloud platforms. This infrastructure operates strictly as a decision-support engine, presenting predictions alongside historical benchmarks rather than claiming direct causal source attribution.

System Integration

The analytics pipeline integrates directly into existing urban environmental workflows by ingesting periodic data updates from public portals and central telemetry networks. New incoming city-year observations are funnelled through identical automated preprocessing blocks, executing median imputation via the internal SimpleImputer pipeline step to prevent any post-hoc data leakage.

The architecture supports an iterative data-ingestion cycle where updated transportation registers, forestry census metrics, and climate logs continuously expand the underlying training matrix. Model performance is automatically reassessed across these expanding windows using the identical **Repeated 5-Fold Cross-Validation (10 repeats)** framework to evaluate and track predictive stability over time before updating production weights.

User Interaction

Planners and policymakers interact with the platform through a unified visual layout. Upon submitting municipal configuration parameters, the user interface displays a multi-layered diagnostic overview:

- **Predicted PM2.5 Level:** Generated by the cross-validated Random Forest core.
- **Localized Feature Footprint:** Rendered via an on-the-fly **SHAP Force Plot** showing exactly which parameters pulled the specific prediction up or down.
- **Global Model Context:** An inline display of the global **SHAP Plot** to reinforce general variable dynamics (e.g., the strong downward pull of expanded canopy cover).

- **Statistical Regression Boundaries:** A side-by-side display of the corrected **HAC (Newey-West) parameter table**, explicitly presenting coefficients and robust p-values.

This interactive design forces users to interpret individual machine learning predictions alongside rigorous statistical verification, preventing them from treating non-linear associations as simple causal proofs.

Benefits to Users

The proposed platform yields immediate operational and strategic planning advantages:

- **Analytical Efficiency:** Replaces isolated bivariate parsing by evaluating urban and climate features inside a single, unified prediction framework.
- **Targeted Planning Support:** Isolates true environmental signals by stripping away intense background meteorological noise, allowing users to accurately evaluate the impact of structural variables.
- **Optimized Resource Allocation:** Identifies specific high-risk city-year conditions—such as dry spells paired with high population density—where the model predicts severe pollution spikes, enabling defensive resource placement.

The financial return is primarily indirect, achieved by optimizing municipal forestry budgets and environmental monitoring assets toward interventions with the highest verified air quality yields.

Example Use Case

A municipal authority can deploy the dashboard to evaluate an upcoming regional expansion project. By inputting projected population density figures, planned vehicle electrification targets, and proposed green space allocations, the planner can generate a future PM2.5 exposure profile.

If the model predicts a severe pollution warning, the integrated **SHAP matrix** identifies whether the spike is driven by a non-linear density threshold or a localized canopy deficit. Rather than assuming the electrification target will automatically neutralize the risk, the authority can use the dashboard to visually confirm that expanding GreenCover provides a much more stable and reliable downward pull on ambient pollution. The results are treated as predictive decision support, not as a causal guarantee.

Limitations and Further Improvements

Limitations

The analytical and predictive scope of this framework is constrained by several structural boundaries:

1. **Sample Constraints:** The underlying dataset contains **191 city-year observations (2017–2025)**, restricting the use of deep architectures and hyper-parameter tuning.
2. **Imputation Noise:** Gaps in meteorological tracking required median imputation within the training pipeline, introducing localized variance into the error profile.
3. **Omitted Confounders:** Crucial primary pollution contributors—such as industrial point-source emissions, active construction dust, biomass burning, and total traffic volumes - could not be consistently tracked across all city-year entries due to data gaps in public reporting.

4. **Observational Design:** The framework uses an observational data matrix that cannot establish mathematical causality or true source attribution.
5. **Predictive vs. Causal Disconnect:** The high predictive importance of tree-based models reflects variable usefulness within an algorithmic partition, not structural statistical causality.

Impact of Limitations

These limitations introduce specific error profiles that require cautious interpretation. Omitted variables can introduce confounding bias into both linear coefficients and tree splits. This is clearly shown by the EV variable, which acts as a proxy for total city scale rather than a pure clean-air asset.

Furthermore, the small sample size exposes standard models to validation volatility. This risk was directly managed by running an extensive **Repeated K-Fold Cross-Validation framework over 50 unique splits** to confirm that the Random Forest model's superior performance ($R^2 = 0.7300$) over the OLS baseline ($R^2 = 0.3420$) was completely stable.

Most importantly, the framework handles the fundamental difference between parametric and non-parametric testing directly: the regression coefficients quantify linear associations under controlled settings, while the **SHAP importances** capture total predictive impact across complex, non-linear spaces.

Future Improvements

The analytical framework can be enhanced through the following direct modifications:

- Expanding the data matrix to integrate a wider array of tier-2 and tier-3 Indian municipalities over a longer time horizon.

- Incorporating regional satellite-derived aerosol optical depth (AOD) tracking to build a proxy layer for missing industrial and construction metrics.
- Upgrading the preprocessing block from standard median imputation to spatial-temporal iterative chained equations (MICE) to minimize residual noise.
- Transitioning the linear framework into longitudinal panel models or quasi-experimental difference-in-differences designs to move closer to causal evaluation.

Future Scope

The long-term objective of this research path is to scale this dual-method framework into a comprehensive, national-level urban air-quality decision-support platform. Future iterations will incorporate look-ahead forecasting horizons, interactive geographic information system (GIS) mapping layers, and real-time sensor API integration. By maintaining this integrated methodology, the platform will continue to deliver advanced machine learning predictions alongside rigorous statistical verification, helping planners craft actionable policies while clearly distinguishing predictive findings from causal claims.

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